Oscar I. Hernandez

About Me

Broadly interested in topology, I'm pursuing a mathematics Ph.D. to study algebraic topology and category theory.

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Website: http://mathemonads.github.io/

Education

Bard College at Simon's Rock 2013–2017
BA, Mathematics and Computer Science
AA, Liberal Arts and Sciences 2015
Advisor: Aaron Williams
Thesis: Constructing Maximum-Rotating de Bruijn Sequences Using Their Necklace Trees

Industry Experience

Computer Engineer	ARKS	2019–
Software Engineer	Independent Contractor	2018–
Associate STEM Teacher	Success Academy Harlem North West Middle School	2017

Academic Experience

Research Intern (Summer 2016), Research Assistant (Spring 2016), Research Intern (Summer 2015), Tutor (2014 – 2017). Bard College at Simon's Rock.

Publications

Currently under development:

1. Oscar Hernandez, Rany Tith. "Toward Compositional Transfer Learning," ongoing development.

2. Aster Greenblatt, Oscar Hernandez, Robert Hearn, Yichao Hou, Hiro Ito, Aaron Williams, and Andrew Winslow. "Spinning Around and Around: Turnstile Puzzles are PSPACE-Complete," under revision.

Refereed papers:

3. Patrick Dragon, Oscar Hernandez, Joe Sawada, Aaron Williams, Dennis Wong. "Constructing de Bruijn sequences with co-lexicographic order: The k-ary Grandmama sequence," European Journal of Combinatorics, 2018.

4. Harold Hastings, Oscar Hernandez, Lucy Jiang, Boqiao Lai, Lindsey Tensen, June Yang. "Dynamics of Biomimetic Electronic Artificial Neural Networks," International Conference on Applications in Nonlinear Dynamics (ICAND), 2016.

5. Patrick Dragon, Oscar Hernandez, and Aaron Williams. "The Grandmama de Bruijn Sequence for Binary Strings," Latin American Theoretical Informatics Symposium (LATIN), 2016.

Workshop papers, invited presentations, and small articles:

6. "Towards Compositional Transfer Learning." Presented at the AMS Fall Western Sectional Meeting, November 2019.
7. "Zero Shot Sentiment Analysis on Tweets in Any Language." Presented at the SoCal NLP Symposium, September 2019.
8. "Automating Art Snobbiness," SigBOVIK 2018, and similar articles every year since 2016.
9. "Analog Artificial Neural Networks." Presented at the Fall 2016 New York State Section of the American Physical Society (NYSS-APS) symposium on the Physics of 2D Materials. Won 2nd place award in the undergraduate research competition.

Teaching

Teacher at Success Academy Charter Schools for 5th grade classes on World History, English Language Arts, Mathematics, Earth Science, and 5-8th grade Computer Science electives.

Subject/Class Tutor at Bard College at Simon's Rock for high school classes on Algebra I with Geometry, Algebra II and Trigonometry, Critical Reasoning, and Computer Science and college classes on Accelerated Beginning Latin I, Introduction to Computer Science, Formal Logic, Vector Calculus, Algorithms & Data Structures, Discrete Mathematics, Graph Theory & Combinatorics.

Awards

1. Intel Edge AI Scholarship (2019)
2. Bertelsmann Technology Scholarship: Artificial Intelligence Track (2019)
3. NSF I-Corps grant awardee x2 (Spring 2019 UCR, Fall 2019 CSUN) for a startup member of the Interphase Incubator's inaugural cohort
4. BA Senior Thesis, awarded Honors (2017)
5. Best Social/Civic Hack (Citrus Hack 2018), Superstar Award & Effective Use of Technology Award (Hack SRC 2017)
6. NYSS-APS undergraduate research competition, 2nd place (2016)
7. Simon's Rock Dean's List (Fall 2015 and Fall 2016)
8. Simon's Rock Supplemental Return Grant (2014–2017)
9. W.E.B. Du Bois Scholarship (2013–2017) and Alumni Scholarship (2013–2017).